

# How to succeed

- Read the Chapter ahead of lectures
- Come to class
- Start paper early
- Study groups
- Prepare exams
- Don't cheat, plagiarize, or otherwise participate in un-ethical behavior
- **Learn with Internet: Wikipedia, google(baidu)...**
  - \*find useful websites for study**
- Ask questions and Think skeptically

# Introduction to Polymer Materials

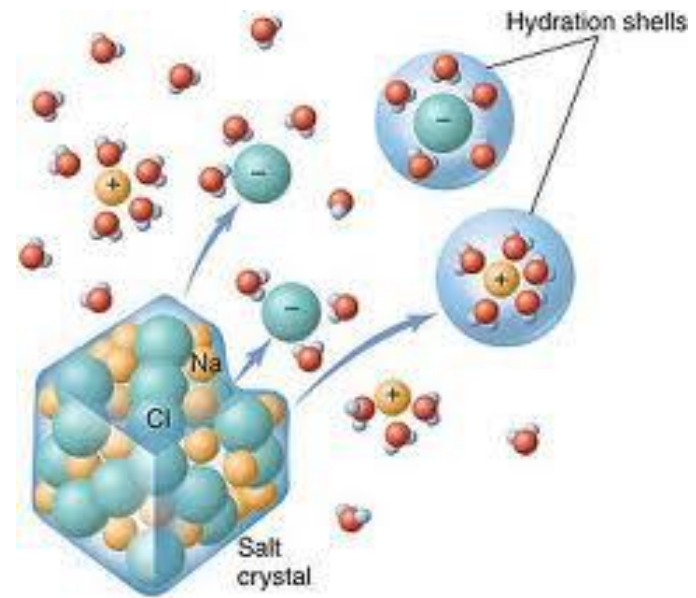
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## Lecture 3: Polymer Solutions

SUN Dazhi

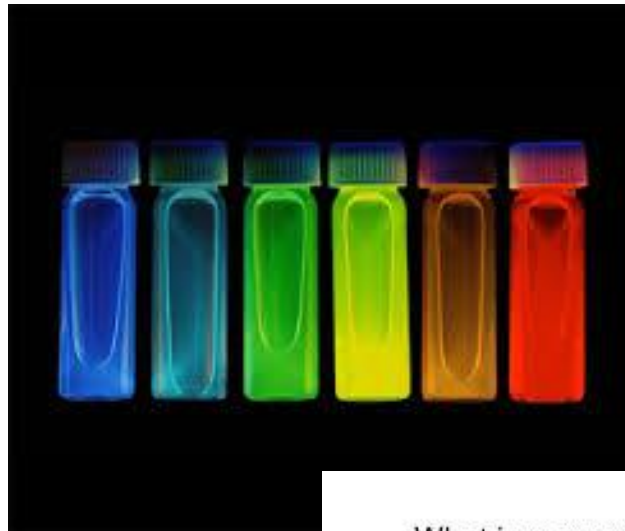
# How do you know if something is soluble???

## Dissolving Salt (NaCl)

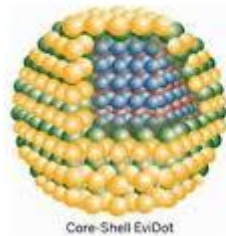


# How do you know if something is soluble???

## Dissolving Nanomaterials (Quantum Dots)



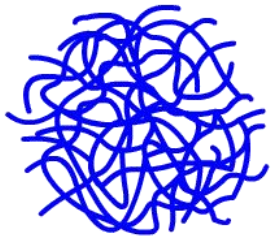
What is a quantum dot?



- Nanocrystals
- 2-10 nm diameter
- semiconductors

# How do you know if something is soluble???

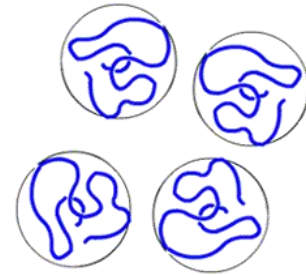
## Dissolving Polymers



a) Polymer molecules in solid state just after being added to a solvent



b) First step: a swollen gel in solvent



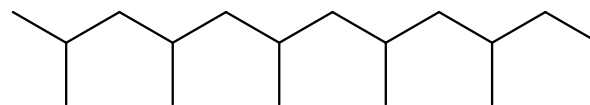
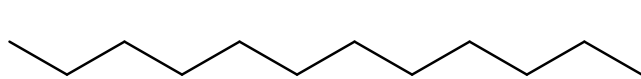
c) Second step: solvated polymer molecules dispersed into a solution

- **Polymer in Solvent**
- **Polymer in Polymer (Polymer Blend/Alloy)**

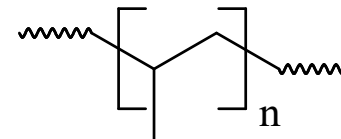
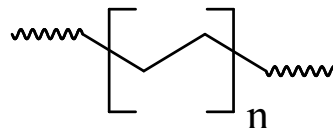
**How about crosslinking/network polymers?**

# Polymer Solubility

- When two hydrocarbons such as dodecane and 2,4,6,8,10-pentamethyldodecane are combined, we generate a homogeneous solution:



- It is therefore interesting that polymeric analogues of these compounds, poly(ethylene) and poly(propylene) do not mix, but when combined, produce a dispersion of one material in the other. **Why ?**



# Applications of Polymer Solutions



**Glues**

**Paintings**

**Wet Spinning Fibers**

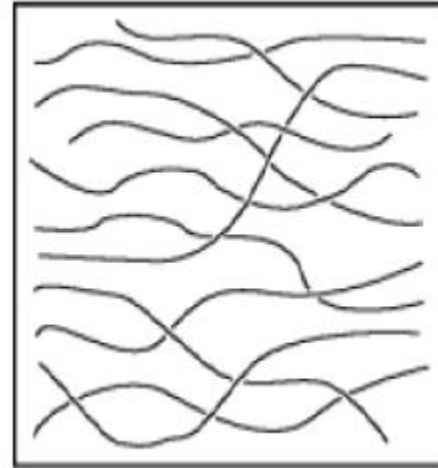
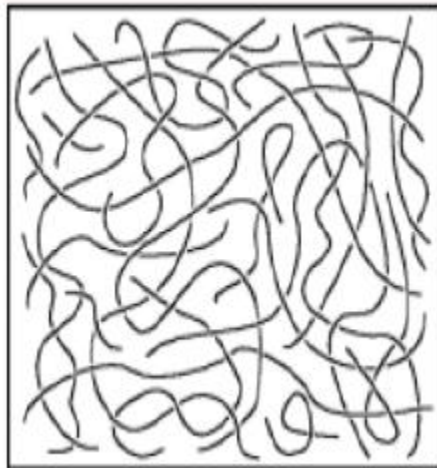
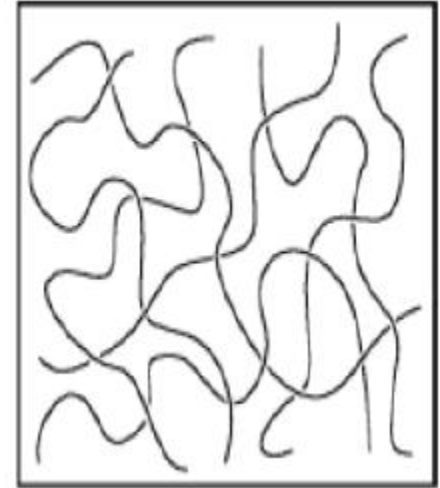
# Applications of Polymer Solutions

**Table 4.1** Selected industrial applications of polymer solutions and precipitates

| Polymer                                  | Solvent                               | Effect                                                                                     | Application                                                                                  |
|------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Sodium carboxymethyl cellulose           | Soapy water                           | Selective precipitation onto clothing fibers                                               | Prevents oils from redepositing on clothing during detergent washing; antiredeposition agent |
| Diblock copolymers                       | Motor oil                             | Colloidal suspensions dissolve at high temperatures, raising viscosity                     | Multiviscosity (constant viscosity) motor oil. Example: 10W-40                               |
| Poly(ethylene oxide)<br>$M = 10^6$ g/mol | Water                                 | Reduces turbulent flow                                                                     | Heat exchange systems, reduces pumping costs                                                 |
| Proteins                                 | Wine                                  | Gels on reacting with tannin                                                               | Clarification of wines, removes colloidal matter                                             |
| Polystyrene, various                     | Triglyceride oils                     | Viscosity control, phase-separates during oil polymerization                               | Oil-based house paints, makes coatings harder, tougher                                       |
| Polyurethanes, various                   | Esters, alcohols, various             | Solvent vehicle evaporates, leaving polymer film for glues, solvent enters mating surfaces | Varnishes, shellac, and glues (adhesives)                                                    |
| Poly(vinyl chloride)                     | Dibutyl phthalate                     | Plasticizes polymer                                                                        | Lower polymer $T_g$ , soften polymer, makes "vinyl"                                          |
| Polystyrene                              | Poly(2,6-dimethyl-1,4-phenyleneoxide) | Mutual solution; toughens polystyrene                                                      | Impact-resistant objects, such as appliances                                                 |
| Poly(methyl methacrylate)                | Poly(vinylidene fluoride)             | Increases PMMA oil and solvent resistance                                                  | Automotive applications, parts that might contact gasoline                                   |

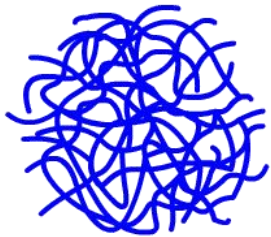


# Polymer Solutions are an ideal model to study polymer chains



# Dissolving Polymers in Solvent

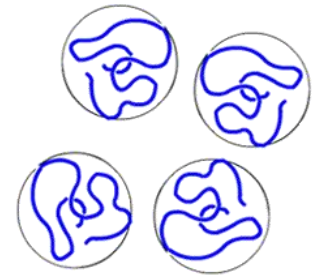
- **first step : the solvent diffuses into polymer masses to make a swollen polymer gel**
- **second step : swollen polymer gel breaks up to solution**



a) Polymer molecules in solid state just after being added to a solvent



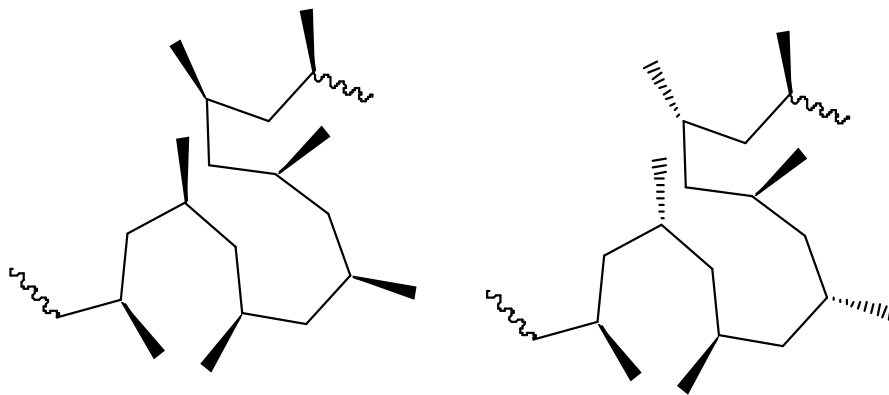
b) First step: a swollen gel in solvent



c) Second step: solvated polymer molecules dispersed into a solution

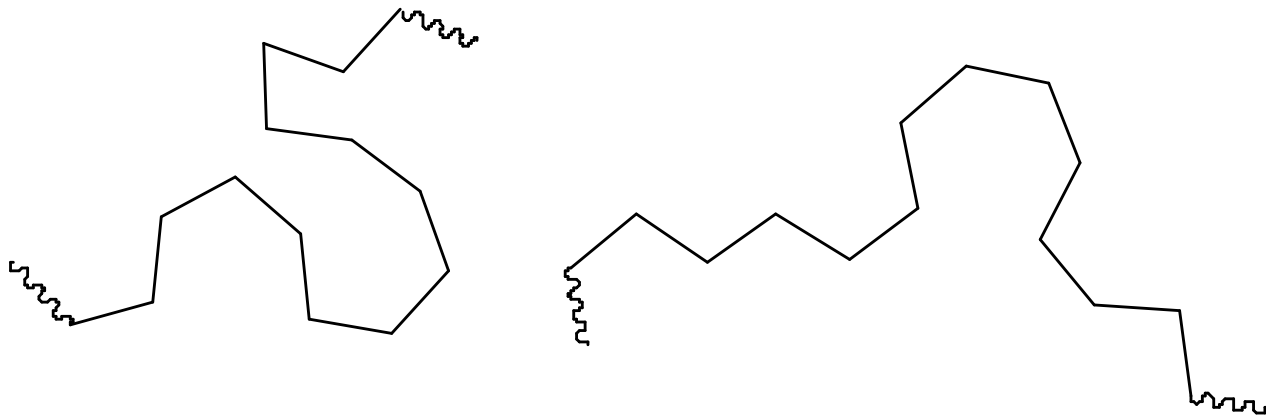
# Polymer Chain

- **Configuration:** arrangement of atoms through bonds
  - Inter-change requires bond breaking and making ( $>200$  kJ/mol)
  - Different compositions of polymers
  - Different isomers or stereochemistries



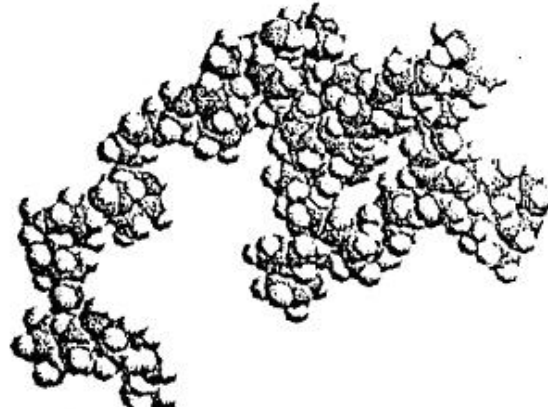
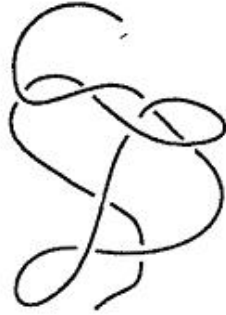
# Polymer Chain

- **Conformation:** within the constraints of a configuration, the possible arrangement(s) of atoms in space
  - Interchange requires bond rotations, but no bonds are broken or made (can have non-bonding , ie. hydrogen bonding)



# Polymer Conformations

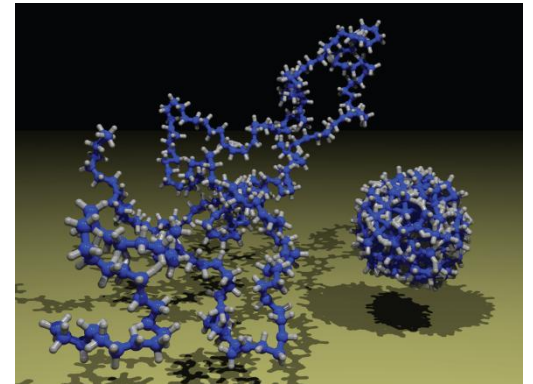
Flexible coil



Rigid rod



And everything in between



# Dimensions of a Polymer Chain

- Dimensions of a polymer chain in solution are affected by:
  - Bonds in polymer:
    - stiffness
  - Chain architecture:
    - molecular weight,
    - branching
  - Environment of the polymer chain
    - solvent

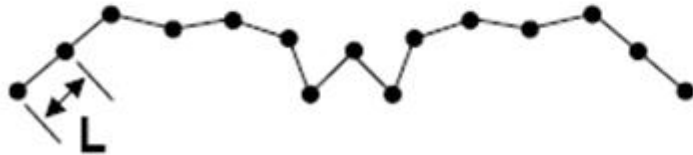


# Contour Length

**Contour length of a polymer chain  
= distance along the chain**



or, at the molecular level:

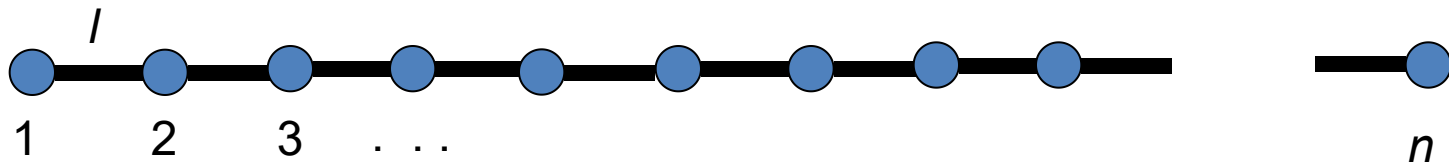


For “n” bonds of length “L”:

**Contour length =  $nL$**

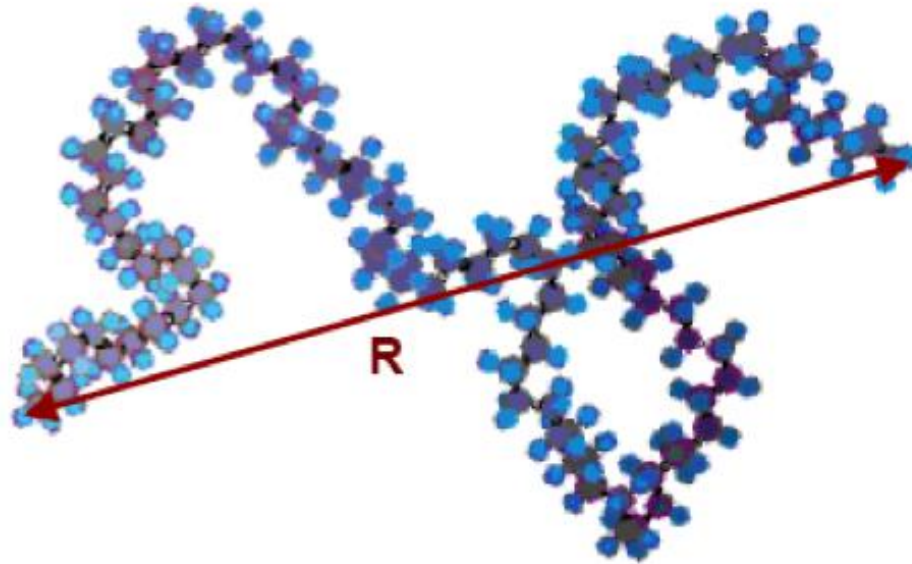
# How to describe conformations with models

- Simplest measure of a chain is the length along the backbone
  - For  $n$  monomers each of length  $l$ , the contour length is  $nl$





# End-to-End Distance $r$

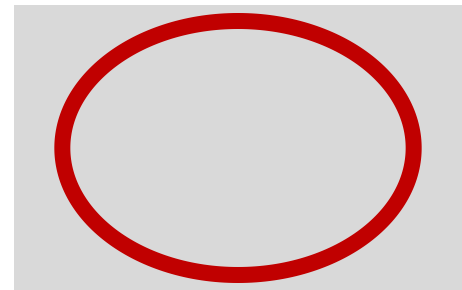


- For an isolated polymer in a solvent the end-to-end distance will change continuously due to molecular motion

# End-to-End Distance $r$

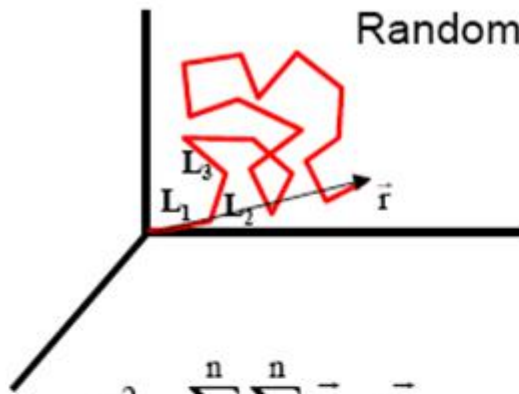


$$R = ?$$



$$R = ?$$

# Freely Jointed Chain: Ideal Polymer



Random walk

$$\vec{r} = \vec{L}_1 + \vec{L}_2 + \vec{L}_3 + \dots + \vec{L}_n$$

$$r^2 = \vec{r} \cdot \vec{r}$$

$$r^2 = (\vec{L}_1 + \vec{L}_2 + \vec{L}_3 + \dots + \vec{L}_n) \cdot (\vec{L}_1 + \vec{L}_2 + \vec{L}_3 + \dots + \vec{L}_n)$$

$$r^2 = \sum_{i=1}^n \sum_{j=1}^n \vec{L}_i \cdot \vec{L}_j$$

For  $i = j$  (happens  $n$  times), the contribution is  $L^2$ .

$$\rightarrow r^2 = nL^2 + 2 \sum_{i=1}^{n-1} \sum_{\substack{j=i+1 \\ i \neq j}}^n \vec{L}_i \cdot \vec{L}_j$$

$= 0$

Now, take average of both sides: **freely-jointed chains.**

$$\langle r^2 \rangle_o = nL^2$$

**Any bond angles  
& orientations  
are possible**

# End-to-End Distance $r$

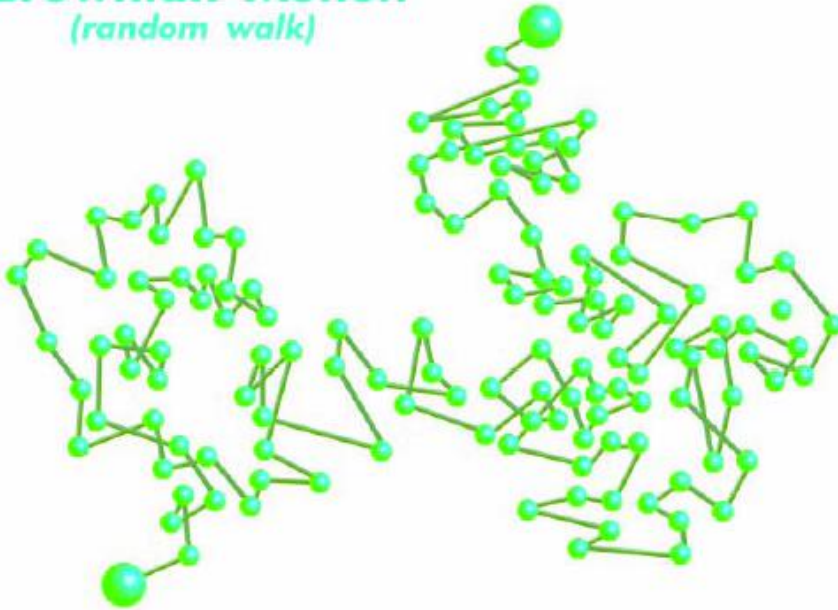
- Define the root mean square (rms) end-to-end distance

$$\left\langle r^2 \right\rangle^{1/2} = ?$$

But we still don't know the shape; need something more

# Random Walk

*Brownian Motion*  
(random walk)



*Redrawn from J. Perrin, Atoms, English translation by  
D. L. Hammick, Constable and Company, London, 1916.*



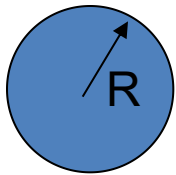
**Jean Perrin**  
**Nobel 1926**

**An ideal polymer chain with  $10^6$  repeat units (not unusual), each unit about  $6\text{\AA}$  will have:**

- contour length = ?**
- rms end-to-end distance  $R$  =?**

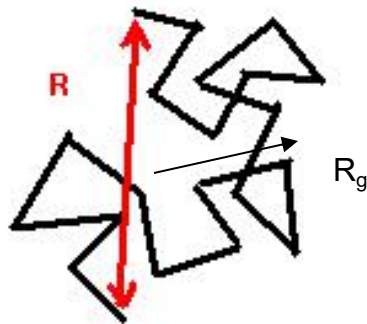
# Radius of Gyration of a Polymer Coil

- The radius of gyration  $R_g$  is defined as the RMS distance of the collection of atoms from their common centre of gravity.



For a solid sphere of radius  $R$ ;

$$R_g = \sqrt{\frac{2}{5}}R = 0.632R$$



For a polymer coil with rms end-to-end distance  $R$  ;

$$R_g = \frac{1}{6} \langle \overline{R}^2 \rangle^{1/2} = \frac{a}{6} N^{1/2}$$

*Can be related directly to statistical distributions from polymer characterizations*

# Radius of Gyration of a Polymer Coil

---

An ideal polymer chain with  $10^6$  repeat units (not unusual), each unit about  $6\text{\AA}$  will have:

- contour length =  $600\text{ }\mu\text{m}$
- rms end-to-end distance  $R = 600\text{ nm}$
- Radius of Gyration  $R_g = ?$

# More Realistic Polymer Chain

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- Real polymer chains are not *freely rotating*
  - Fixed bond angle
  - Restricted rotation angle(steric)
  - Excluded volumes



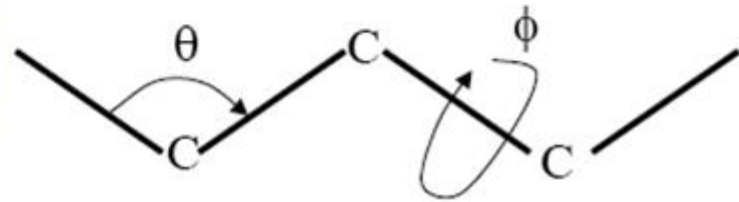
# Valence Angle Model

Real chains are not *freely jointed* with unlimited bond angles.

$$\rightarrow \langle r^2 \rangle_o = nL^2 + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n \langle \vec{L}_i \cdot \vec{L}_j \rangle$$

$$\uparrow \langle |\vec{L}_i| |\vec{L}_j| \cos \theta_{ij} \rangle$$

$$\langle r^2 \rangle_o = nL^2 + 2L^2 \sum \sum \langle \cos \theta_{ij} \rangle$$



For fixed  $\theta$  and large  $n$ ,

$$\langle r^2 \rangle_o = nL^2 + 2L^2 \sum \sum \langle \cos \theta_{ij} \rangle$$



$$\langle r^2 \rangle_o = nL^2 \left( \frac{1 - \cos \theta}{1 + \cos \theta} \right)$$

Fixed  $\theta$ ,  
Free  $\phi$ .

# Valence Angle Model

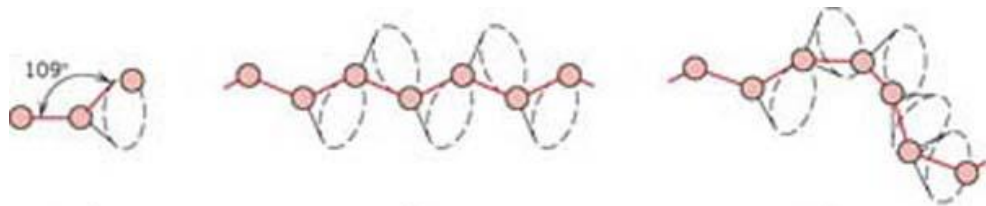
- Simplest modification to the freely jointed chain model
  - Introduce bond angle restrictions
  - Allow free rotation about bonds
  - *Neglecting* steric effects (for now)

- If all bond angles are equal to  $\theta$ ,

$$\langle r^2 \rangle_{fa} = nl^2 \frac{(1 - \cos \theta)}{(1 + \cos \theta)}$$

- E.g. for polyethylene  $\theta = 109.5^\circ$  and  $\cos \theta \sim -1/3$ , hence,

$$\langle r^2 \rangle_{fa} \approx 2nl^2$$



Ex. Polyethylene  $n \sim 10,000$

Freely Jointed

$$\langle r^2 \rangle^{1/2} = n^{1/2} L$$

$$\langle r^2 \rangle^{1/2} = (10000)^{1/2} \cdot 1.54 \text{ \AA} = 150 \text{ \AA}$$

Straight

$$r = (10000) \cdot 1.54 \text{ \AA}$$

$$r = 15,400 \text{ \AA}$$

Valence Angle Model ( $\theta = 109.5^\circ$ )

$$\langle r^2 \rangle_o = n L^2 \left( \frac{1 - \cos \theta}{1 + \cos \theta} \right) = 2n L^2$$

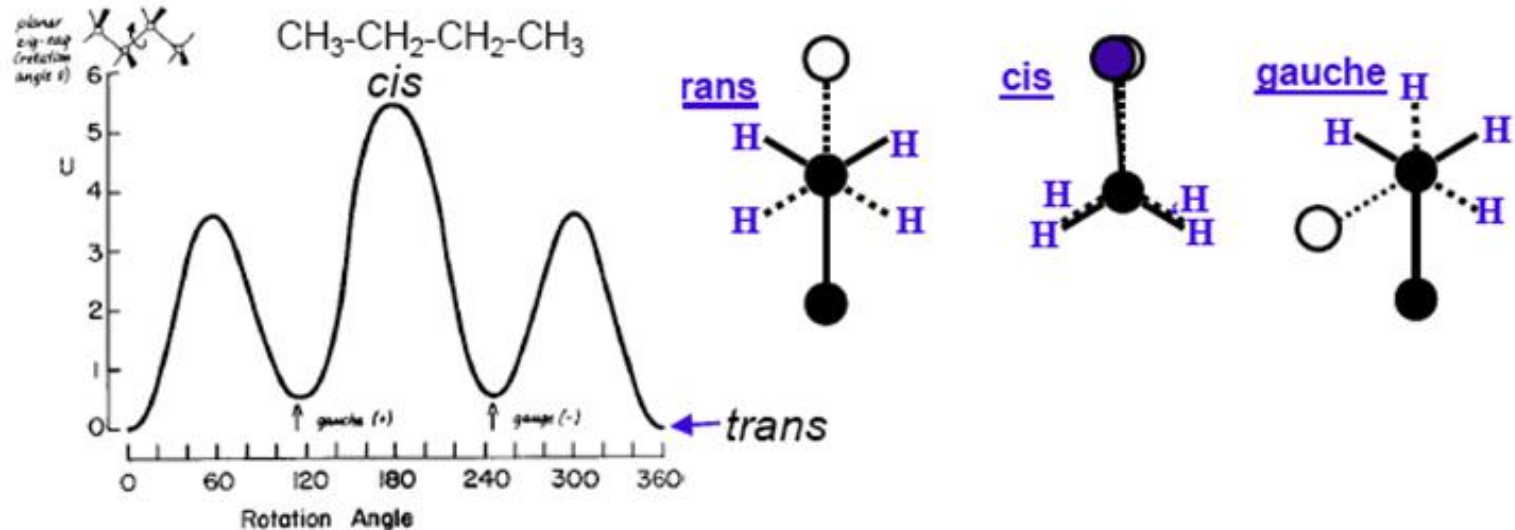
$$\langle r^2 \rangle^{1/2} = \sqrt{2} \times \sqrt{n} L = 212.1 \text{ \AA}$$

But what about the *rms* radius of gyration?

$$\langle s^2 \rangle_o^{1/2} = \frac{\langle r^2 \rangle_o^{1/2}}{\sqrt{6}} = \frac{150}{\sqrt{6}} \text{ \AA} = 61.2 \text{ \AA}$$

# Restricted Rotation Angle

Real chains possess *restricted rotations*.



Accounting for fixed bond angles **and restricted rotations**:

$$\langle r^2 \rangle_o = nL^2 \left( \frac{1 - \cos \theta}{1 + \cos \theta} \right) \left( \frac{1 + \langle \cos \phi \rangle}{1 - \langle \cos \phi \rangle} \right)$$

# Steric parameter and the characteristic ratio

Real chains are not *freely rotating*.

$$\langle r^2 \rangle_o = nL^2 \left( \frac{1 - \cos \theta}{1 + \cos \theta} \right) \left( \frac{1 + \langle \cos \phi \rangle}{1 - \langle \cos \phi \rangle} \right) \quad \Rightarrow \quad \langle r^2 \rangle_o = \sigma^2 nL^2 \left( \frac{1 - \cos \theta}{1 + \cos \theta} \right)$$

$\sigma$  is the steric parameter

**Characteristic Ratio**

$$C_\infty \equiv \frac{\langle r^2 \rangle}{nL^2} \quad 5 < C_\infty < 10 \text{ for most polymers}$$

$C_\infty$

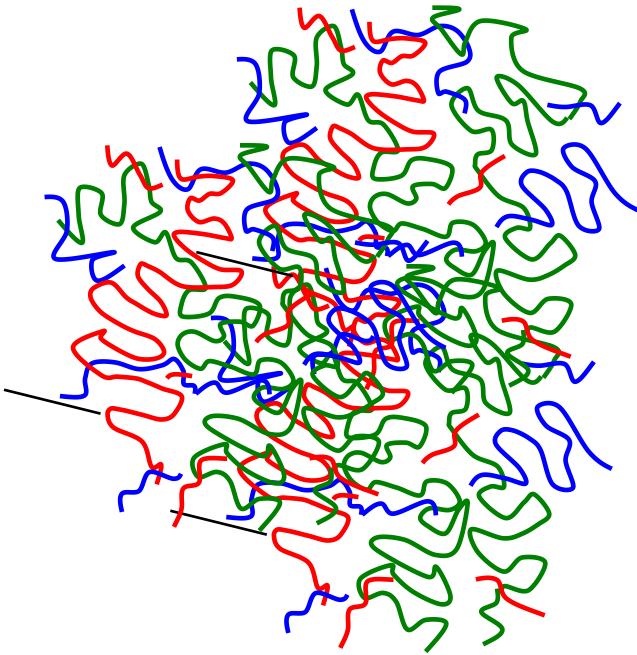
| Polymer                   | Temperature/°C | $\sigma$ | $C_\infty$ |
|---------------------------|----------------|----------|------------|
| Polyethylene              | 140            | 1.8      | 6.8        |
| Isotactic polypropylene   | 140            | 1.6      | 5.2        |
| Poly(vinyl chloride)      | 25             | 1.8      | 6.7        |
| Polystyrene               | 25             | 2.3      | 10.8       |
| Polystyrene               | 70             | 2.1      | 9.2        |
| Poly(methyl methacrylate) | 25             | 2.1      | 8.6        |
| Poly(methyl methacrylate) | 72             | 1.8      | 6.6        |

$C_\infty$  is a function of the stiffness of the chain. Higher is stiffer

# Excluded Volume

- Freely jointed chain, valence angle and rotational isomeric states models all ignore
  - long range intramolecular interactions (e.g. ionic polymers)
  - polymer-solvent interactions
- Such interactions will affect  $\langle r^2 \rangle$ 
  - Define  $\langle r^2 \rangle = \alpha_r^2 \langle r^2 \rangle_0$   
where  $\alpha_r$  is the expansion parameter

# Space Filling



- The random walk and the steric limitations makes the polymer coils in a polymer melt or in a polymer glass “expanded”.
- However, the overlap between molecules ensure space filling

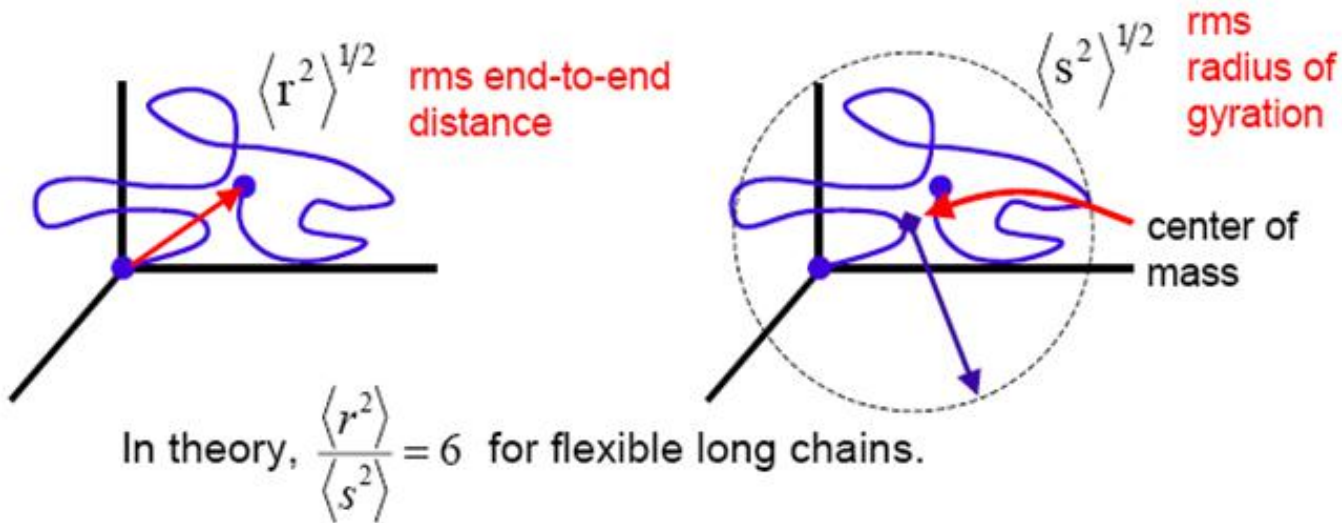
# Polymer Chain in Solution

Flexible polymers assume coiled conformations in solution and in bulk in order to increase their entropy.

$$S = k \ln \Omega$$



Entropy decreases because the number of different conformations ( $\Omega$ ) decreases.





# Expansion Factors

$\overline{r}^2$  : mean-square average distance

$\overline{s}^2$  : square average radius of gyration

**Better solvent → stronger interaction between solvent and polymer → larger hydrodynamic volume**

$$\overline{r}^2 = r_o^2 \alpha^2$$

$$\overline{s}^2 = s_o^2 \alpha^2$$

$$\overline{r}^2 = 6\overline{s}^2$$

$r_o, s_o$ : **unperturbed dimension** (i.e. the size of the polymer exclusive of solvent effects)

$\alpha$  : **expansion factor**

(for a linear polymer)

# Expansion Factors

$$\alpha = \frac{(\overline{r^2})^{\frac{1}{2}}}{(\overline{r_o^2})^{\frac{1}{2}}}$$

greater  $\alpha \rightarrow$  better solvent  
 $\alpha = 1 \rightarrow$  ideal statistical coil

**Solubility vary with temperature** in a given solvent  
 $\rightarrow \alpha$  is temperature dependent

# Expansion Factors

- **Theta ( $\theta$ ) temperature (Flory temperature)**
  - For a given polymer in a given solvent, **the lowest temperature at which  $\alpha = 1$**
  - $\theta$  state /  $\theta$  solvent

$$\alpha = \frac{(\bar{r}^2)^{\frac{1}{2}}}{(\bar{r}_o^2)^{\frac{1}{2}}}$$

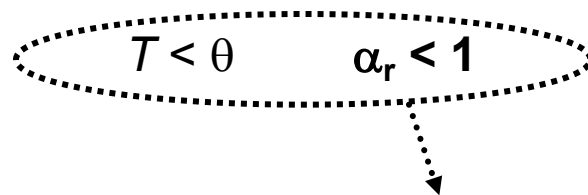
- **Polymer in a  $\theta$  state**
  - having a **minimal solvation effect**
  - on the border of becoming insoluble
  - further diminution of solvation effect → polymer precipitation

# Expansion Factors

- $a_r$  depends on balance between polymer-solvent and polymer-polymer interactions
  - If *polymer-polymer* are *more* favourable than *polymer-solvent*
    - $a_r < 1$
    - Chains contract
    - Solvent is poor
  - If *polymer-polymer* are *less* favourable than *polymer-solvent*
    - $a_r > 1$
    - Chains expand
    - Solvent is good
  - If these interactions are equivalent, we have theta condition
    - $a_r = 1$
    - Same as in amorphous melt

# The Theta Temperature

- For most polymer solutions  $\alpha_r$  depends on temperature, and increases with increasing temperature
- At temperatures above some *theta temperature*, the solvent is good, whereas below the solvent is poor, i.e.,

$$\begin{array}{ll} T > \theta & \alpha_r > 1 \\ T = \theta & \alpha_r = 1 \\ T < \theta & \alpha_r < 1 \end{array}$$


**Polymers will precipitate out of solution**

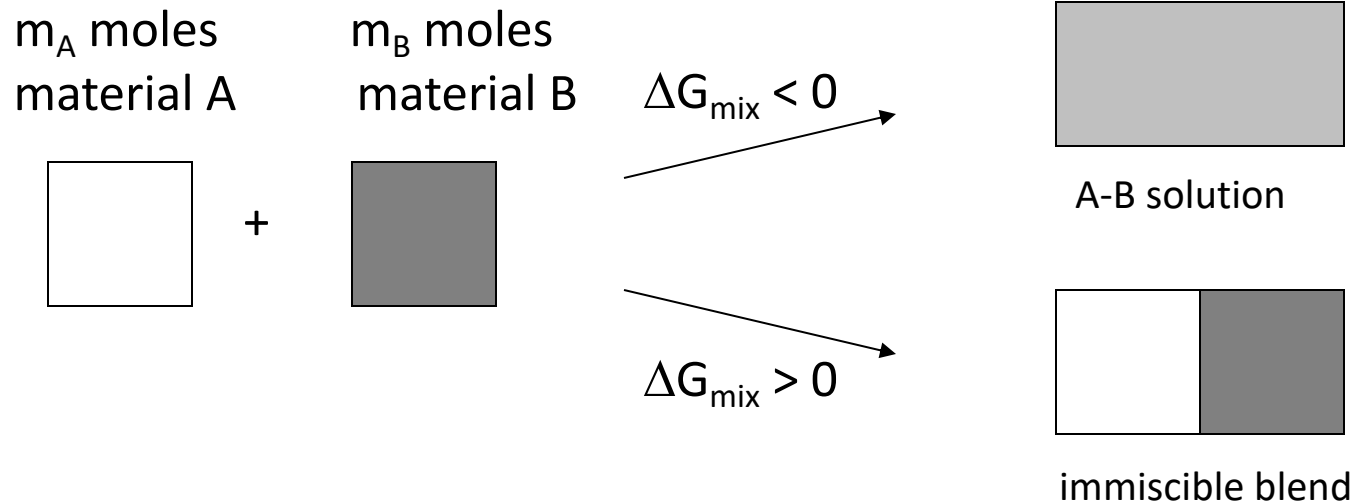
# Solubility of Polymers

## Encyclopedia of Polymer Science and Technology

**A polymer is often soluble in a low molecular weight liquid if:**

- the two components are similar chemically or are so constituted that specific attractive interactions such as hydrogen bonding take place between them;
- the molecular weight of the polymer is low;
- the bulk polymer is not crystalline;
- the temperature is elevated (except in systems with LCST).

# Thermodynamics of Mixing



$\Delta G_{\text{mix}}$  (Joules/gram) is defined by:

$$\Delta G_{\text{mix}} = \Delta H_{\text{mix}} - T \Delta S_{\text{mix}}$$

where  $\Delta H_{\text{mix}} = H_{AB} - (x_A H_A + x_B H_B)$

$$\Delta S_{\text{mix}} = S_{AB} - (x_A S_A + x_B S_B)$$

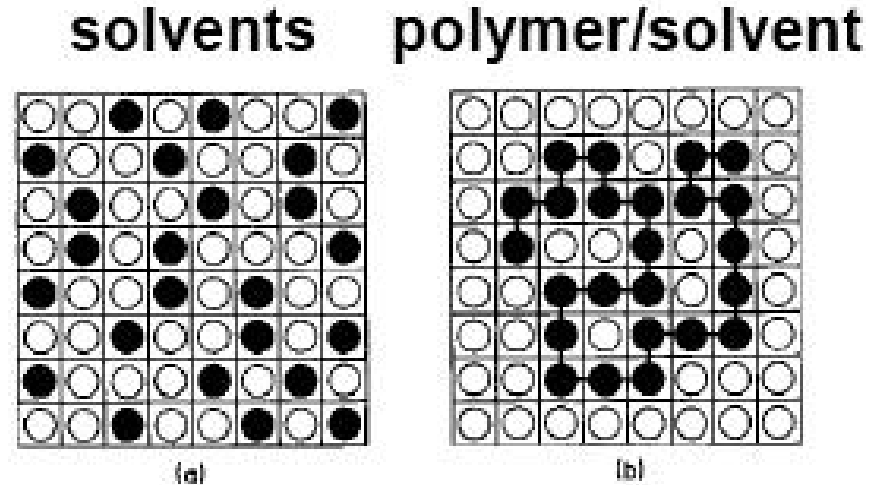
and  $x_A$ ,  $x_B$  are the mole fractions of each material.

***Polymer-solvent: volume fraction of polymers***

# Solvent-Solvent Solutions vs Polymer-Solvent Solutions

## Mixing Free Energy

$$\Delta G_m = \Delta H_m - T\Delta S_m$$



- Solvents can easily replace one another.
  - Polymers are thousands of solvent sized monomers connected together.
  - Monomers can not be placed randomly!
- 
- *Entropy of mixing is generally positive*
  - *Enthalpy is also often positive: smaller the better, Ideal = 0*



# Enthalpy of Mixing

$\Delta H_{\text{mix}}$  can be a positive or negative quantity

- If A-A and B-B interactions are stronger than A-B interactions, then  $\Delta H_{\text{mix}} > 0$  (unmixed state is lower in energy)
- If A-B interactions are stronger than pure component interactions, then  $\Delta H_{\text{mix}} < 0$  (solution state is lower in energy)

An ideal solution is defined as one in which the interactions between all components are equivalent. As a result,

$$\Delta H_{\text{mix}} = H_{\text{AB}} - (w_{\text{A}}H_{\text{A}} + w_{\text{B}}H_{\text{B}}) = 0 \quad \text{for an ideal mixture}$$

In general, **most polymer-solvent interactions produce  $\Delta H_{\text{mix}} > 0$** , the exceptional cases being those in which significant hydrogen bonding between components is possible.

# ENTROPY OF MIXING

The total configurational entropy of mixing (J/K) created in forming a solution from  $n_1$  moles of solvent and  $n_2$  moles of polymer is:

$$\Delta S_{\text{mix}} = -R(n_1 \ln \phi_1 + n_2 \ln \phi_2)$$

where  $\phi_i$  is the component volume fraction in the mixture:

$$\phi_1 = \frac{x_1 n_1}{x_1 n_1 + x_2 n_2} \quad \text{and} \quad \phi_2 = \frac{x_2 n_2}{x_1 n_1 + x_2 n_2}$$

$x_i$  represents the number of segments in the species

- for a usual monomeric solvent,  $x_i = 1$
- $x_i$  for a polymer corresponds roughly (but not exactly) to the repeat unit

# Flory-Huggins Theory

- Combining expressions for the enthalpy and entropy of mixing generates the free energy of mixing:

$$\begin{aligned}\Delta G_{\text{mix}} &= \Delta H_{\text{mix}} - T\Delta S_{\text{mix}} \\ &= RT\chi\phi_2n_1x_1 + RT(n_1\ln\phi_1 + n_2\ln\phi_2) \\ &= RT(n_1\ln\phi_1 + n_2\ln\phi_2 + \chi\phi_2n_1x_1)\end{aligned}$$

Note that for complete miscibility over all concentrations,  $\chi$  for the solute-solvent pair at the T of interest must be less than 0.5.

- If  $\chi > 0.5$ , then  $\Delta G_{\text{mix}} > 0$  and phase separation occurs
- If  $\chi < 0.5$ , then  $\Delta G_{\text{mix}} < 0$  over the whole composition range.
- The temperature at which  $\chi = 0.5$  is the theta temperature.

# Paul J. Flory

## The Nobel Prize in Chemistry 1974



***"for his fundamental achievements, both theoretical and experimental, in the physical chemistry of the macromolecules"***

# Flory-Huggins Theory

|                                           |                                           |                  |
|-------------------------------------------|-------------------------------------------|------------------|
| $\overline{\Delta G}_1^E > 0 \Rightarrow$ | $\chi > 1/2$ and $T < \theta \Rightarrow$ | poor solvent     |
| $\overline{\Delta G}_1^E = 0 \Rightarrow$ | $\chi = 1/2$ and $T = \theta \Rightarrow$ | $\theta$ solvent |
| $\overline{\Delta G}_1^E < 0 \Rightarrow$ | $\chi < 1/2$ and $T > \theta \Rightarrow$ | good solvent     |

1. At  $\theta$  temperature, the dilute polymer solution is NOT an ideal solution.  
But,  $\overline{\Delta H}_1^E$  and  $\overline{\Delta S}_1^E$  cancel each other!
2. Choosing different solvent and temperature, we can satisfy  $\overline{\Delta G}_1^E = 0$ .  
We say the dilute solution is in  $\theta$  condition.

TABLE 3.1 Values of theta temperature for different polymer-solvent systems\*

| Polymer                   | Solvent        | $\theta/^\circ\text{C}$ |
|---------------------------|----------------|-------------------------|
| Polyethylene              | Biphenyl       | 125                     |
| Polystyrene               | Cyclohexane    | 34                      |
| Poly(vinyl acetate)       | Methanol       | 6                       |
| Poly(methyl methacrylate) | Butyl Acetate  | -20                     |
| Poly(methyl methacrylate) | Pentyl Acetate | 41                      |
| Poly(vinyl alcohol)       | Water          | 97                      |
| Poly(acrylic acid)        | 1,4-Dioxan     | 29                      |

\* The entropy parameter  $\psi$  is positive for all of the systems except poly(vinyl alcohol); water and poly(acrylic acid); 1,4-dioxan for which it is negative.

# $\Delta H_{\text{mix}}$ and the Solubility Parameter

- The most popular predictor of polymer solubility is the solubility parameter,  $\delta_i$ .
- For regular solutions in which intermolecular attractions are minimal,  $\Delta H_{\text{mix}}$  can be estimated through:

$$\Delta H_{1,2} \approx \Delta U_{1,2} = \phi_1 \phi_2 (\delta_1 - \delta_2)^2 \quad \text{cal/cm}^3$$

where  $\Delta U_{1,2}$  = internal energy change of mixing per unit volume,

$\phi_i$  = volume fraction of component i in the proposed mixture,

$\delta_i$  = solubility parameter of component i:  $(\text{cal/cm}^3)^{1/2}$

# Solubility Parameters of Select Materials

**Table 2. Solubilities<sup>a</sup> and Solubility Parameters<sup>b</sup> of Polymer–Solvent Systems<sup>c</sup>**

| Name                 | $\delta_1$ | Polyisobutylene<br>( $\delta_2 = 16.2$ ) | Poly(methyl<br>methacrylate)<br>( $\delta_2 = 18.6$ ) | Poly(vinyl<br>acetate)<br>( $\delta_2 = 19.2$ ) | Poly(hexamethylene<br>adipamide)<br>( $\delta_2 = 27.8$ ) |
|----------------------|------------|------------------------------------------|-------------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------|
| decafluorobutane     | 10.6       | —                                        | —                                                     | —                                               | —                                                         |
| neopentane           | 12.9       | +                                        | —                                                     | —                                               | —                                                         |
| <i>n</i> -hexane     | 14.9       | +                                        | —                                                     | —                                               | —                                                         |
| diethyl ether        | 15.1       | —                                        | —                                                     | —                                               | —                                                         |
| cyclohexane          | 16.8       | +                                        | —                                                     | —                                               | —                                                         |
| carbon tetrachloride | 17.6       | +                                        | +                                                     | —                                               | —                                                         |
| benzene              | 18.8       | +                                        | +                                                     | —                                               | —                                                         |
| chloroform           | 19.0       | +                                        | +                                                     | +                                               | —                                                         |
| methyl ethyl ketone  | 19.0       | —                                        | +                                                     | +                                               | —                                                         |
| acetone              | 20.3       | —                                        | +                                                     | +                                               | —                                                         |
| carbon disulfide     | 20.5       | —                                        | —                                                     | —                                               | —                                                         |
| 1,4-dioxane          | 20.5       | —                                        | +                                                     | +                                               | —                                                         |
| dimethylformamide    | 24.8       | —                                        | +                                                     | +                                               | (+)                                                       |
| <i>m</i> -cresol     | 27.2       | —                                        | +                                                     | +                                               | +                                                         |
| formic acid          | 27.6       | —                                        | +                                                     | —                                               | +                                                         |
| methanol             | 29.7       | —                                        | —                                                     | —                                               | —                                                         |
| water                | 47.9       | —                                        | —                                                     | —                                               | —                                                         |

<sup>a</sup> + soluble, — insoluble, (+) soluble at high temperatures only.

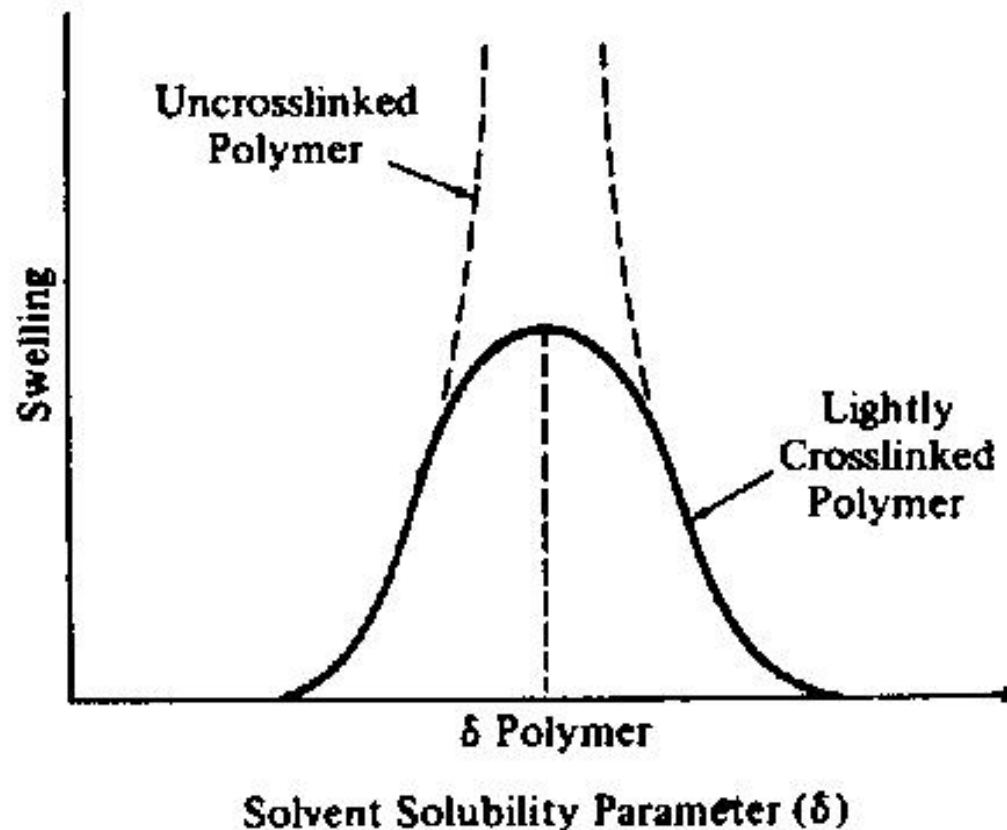
<sup>b</sup> The solubility parameters in this table have units of (J/cm<sup>3</sup>)<sup>1/2</sup>. To convert to (cal/cm<sup>3</sup>)<sup>1/2</sup>, divide by 2.05.

# Solubility Parameter

The conditions of greatest polymer solubility exist when the solubility parameters of polymer and solvent match.

➤ If the polymer is crosslinked, it cannot dissolve but only swell as solvent penetrates the material.

The solubility parameter of a polymer is therefore determined by exposing it to different solvents, and observing the  $\delta$  at which swelling is maximized.





# Factors Influencing Polymer-Solvent Miscibility

Based on the Flory-Huggins theory, we can explain the influence of the following variables on miscibility:

- 1. Temperature:** The sign of  $\Delta G_{\text{mix}}$  is determined by the Flory-Huggins interaction parameter,  $\chi$ .  
As temperature rises,  $\chi$  decreases thereby improving solubility.
- 2. Molecular Weight:** Increasing molecular weight reduces the configurational entropy of mixing, thereby reducing solubility.
- 3. Crystallinity:** A semi-crystalline polymer has a more positive  $\Delta H_{\text{mix}} = H_{\text{AB}} - x_{\text{A}}H_{\text{A}} - x_{\text{B}}H_{\text{B}}$  due to the heat of fusion that is lost upon mixing.

# Keywords in This Lecture

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- **Conformation and Configuration**
- **Polymer coil**
- **End-to-end distance**
- **Radius of Gyration**
- **Ideal polymer chain/fixed bond angle, steric**
- **Expansion factor**
- **Theta temperature**
- **F-H theory**
- **Solubility parameter**

# HOMework

- Describe, **in your own words**, the conformation change of ideal polymers in solution upon decreasing temperature (higher than  $\theta \rightarrow$  close to  $\theta \rightarrow \theta \rightarrow$  slightly lower than  $\theta \rightarrow$  lower than  $\theta$  ).
- \* May use cartoons(drawn by yourself) for better illustration.